

NADA MOHAMED

AI/ML Engineer • Blockchain Engineer

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SUMMARY

Computer Science graduate (2026, Honours) working at the intersection of machine learning and distributed ledgers. Fine-tuned LLMs (DeepSeek-Coder, FLAN-UL2) for a multimodal agent, shipped deep-learning and computer-vision systems (First Place, AASTMT), and engineered Hyperledger Fabric chaincode in Go for a regulated blockchain payroll platform validated by 91 passing smart-contract tests.

EDUCATION

Arab Academy for Science, Technology & Maritime Transport Sep 2022 – 2026
BSc Computer Science (Honours), dual degree with the University of Northampton (UK) – GPA: 3.69 / 4.0 Cairo, Egypt
• Coursework: Machine Learning, Computer Vision, NLP, Algorithms & Optimization, Parallel Computing, Databases.

TECHNICAL SKILLS

AI & Machine Learning: PyTorch, TensorFlow, Keras, HuggingFace Transformers, Scikit-learn, OpenCV, Pandas; LLM fine-tuning, agent orchestration, NLP, Computer Vision, Reinforcement Learning, Explainable AI

Blockchain & Security: Hyperledger Fabric, Chaincode (Go), RAFT ordering, endorsement policies, Privy MPC / threshold custody, non-custodial wallets, X.509 PKI, Cryptography, STRIDE threat modelling

Languages: Python, Go, TypeScript, JavaScript, C/C++, Java, SQL, Bash

Serving & Infrastructure: FastAPI, Flask, React, Docker, Kubernetes, AWS RDS, MySQL, Git, CI/CD

EXPERIENCE

SAIB Bank Aug 2026 – Present
FinTech Intern Cairo, Egypt

- Studying fintech under the Head of FinTech & Startup Partnerships, closing with an emerging-tech case study.

DEPI – Microsoft ML Engineering Curriculum Nov 2024 – May 2025
AI/ML Engineer Intern Zamalek, Egypt

- Completed a 7-month program in NLP, generative AI, computer vision, and model deployment; delivered milestone projects and an end-to-end capstone (Jarvis AI Assistant) with agent orchestration.

Dell Technologies Aug 2025 – Sept 2025
Summer Intern Cairo, Egypt

- Workshops on Docker, Kubernetes, and software engineering; led the Academy Project from concept to proposal.

PROJECTS

Payramid: Blockchain Payroll Microservice | *Go, Hyperledger Fabric, Privy MPC, Twilio, TypeScript* 2025 – 2026

- *Graduation project.* Built a blockchain payroll and immutable-audit layer for instant salary disbursement with cryptographically verifiable proofs of payment, settling on Egypt's national rails (InstaPay, Meeza, ACH) under banking and data-protection law.
- Engineered Hyperledger Fabric chaincode (Go) on a 3-organization RAFT network with MAJORITY endorsement; 91 passing automated tests cover serialization integrity, double-spend prevention, and a STRIDE threat model.
- Delivered non-custodial wallets via Privy threshold (MPC) key management and an encrypted SMS relay with one-time withdrawal codes for feature phones; authored the IEEE 830 SRS and IEEE 829 test plan.

Jarvis AI Assistant 🧠 | *Python, DeepSeek-Coder, FLAN-UL2, HuggingFace Transformers, Whisper* 2025 – Present

- Fine-tuned DeepSeek-Coder and FLAN-UL2 for multimodal code generation, image captioning, and voice interaction.
- Integrated Whisper speech recognition and agent orchestration with custom toolchains for autonomous task execution.

Music Genre Detection 🧠 | *Python, TensorFlow/Keras, Librosa, Flask*

- Designed a CNN-LSTM-Attention model classifying mel-spectrograms into 10 genres, with SHAP/LIME/Grad-CAM explainability.

Oribu: AI Facial Recognition 🧠 | *Python, OpenCV, Siamese Neural Network, Twilio API* 2023

- Built real-time one-shot face verification with a Siamese network and contrastive loss. (*First Place, AASTMT.*)

CERTIFICATIONS, AWARDS & LANGUAGES

🏆 **First Place, AI/ML Project Submission – AASTMT** 🌟 **Microsoft ML Engineer Course – DEPI**

🌟 **AI & Machine Learning Course – Universitat Autònoma de Barcelona** 🏠 **CS Camp – Carnegie Mellon University**

🗣️ **Languages:** Arabic (Native), English (Fluent), Spanish (Intermediate)